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Asymmetric Numeral System

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Outline

Introduction

Introduction

- Hi there, this is a self inspired latex beamer template for Leipzig university.

Entropy Coding

Entropy Coding

- The theoretical limit of compression is given by Shannon entropy, which measures the minimum average number of bits needed to encode a symbol from a data source.
- Lossless compression algorithms aim to approach this limit but cannot go below it.
- The entropy depends on the probability distribution of the symbols, with more skewed distributions being more compressible.
- Shannon Entropy [1]

$$H(X) = -\sum_{i=1}^{n} p(x_i) \log_2 p(x_i)$$

- Shannon's Source Coding Theorem [1]

$$N \cdot H(X)$$

Asymmetric Numeral System

Motivation

ANS combines the compression ratio of arithmetic coding (which uses a nearly accurate probability distribution), with a processing cost similar to that of Huffman coding [4].

Basic Idea

- Symbol spread function

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- Representation of information
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- Representation of information
  (represents an entire message as a single number, similar to Arithmetic Coding) –> example: representation of {2,0,2,5,1,8}
- Symmetric or Asymmetric
- rANS and tANS

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- the function that determines the split of the $\mathbb{N}$ subsets corresponding to different symbols [1]

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Symbol spread function

- the function that determines the split of the $\mathbb{N}$ subsets corresponding to different symbols [1]
- $\bar{s} : \mathbb{N} \to \mathcal{A}, \bar{s}(x) = \text{mod}(x, b) = s$
- In the standard binary system example: $\bar{s}(x) = \text{mod}(x, 2)$
  e.g. splits natural numbers into subsets of even/odd numbers [2]

Symbol spread function (Exa. 1)

$S = \{0, 1\}$

Exa. 1: Standard Binary System

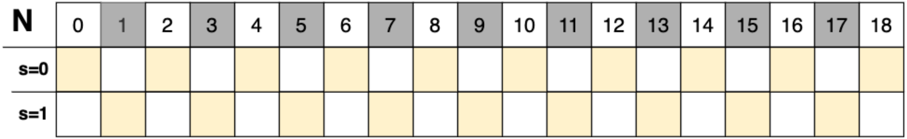

Fig.1

Symbol spread function (Exa. 2)

$S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

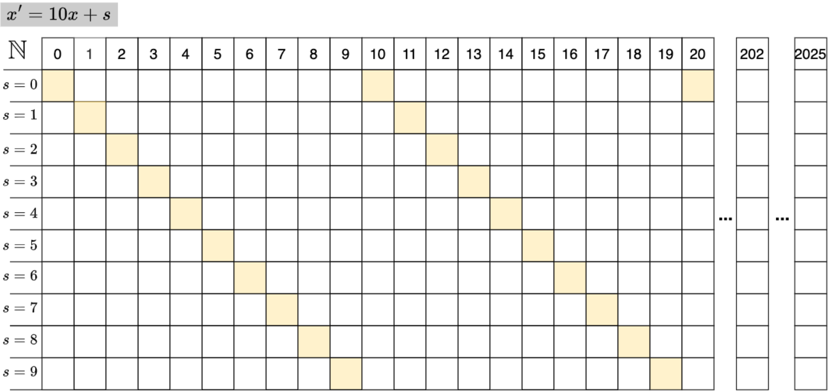

Fig.2

Representation of information (Exa. 2)

- a finite sequence of symbols/digits:

$$\mathcal{A} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

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- the sequence of symbols/digits to be encoded:

$$S_{in} = (2, 0, 2, 5)$$

- How can we represent this sequence as a single number?
- The easiest way is to concatenate the symbols/digits:

$$x_{out} = 2025$$

Representation of Information (Exa. 2)

- encoding the sequence of digits (2,0,2,5,1,8)

$$
\begin{aligned}
x_{\text{init}} &= 0 \\
x_1 &= x_{\text{init}} \times 10 + 2 = 2 \\
x_2 &= x_1 \times 10 + 0 = 20 \\
x_3 &= x_2 \times 10 + 2 = 202 \\
x_{\text{out}} &= x_3 \times 10 + 5 = 2025
\end{aligned}
$$

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$$x_{i+1} = x_i \times 10 + s_i$$

- x’ is the x-th appearance of symbol s

$$x' = C(s, x) = x \cdot 10 + s$$

Representation of Information (Exa. 2)

$x' = 10x + s$

$s = 2$ $s = 0$ $s = 2$ $s = 5$

$\mathbb{N}$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	...	202	...	2025
$s = 0$	0										1										2				
$s = 1$		0										1													
$s = 2$			0										1												
$s = 3$				0										1											
$s = 4$					0										1										
$s = 5$						0										1									
$s = 6$							0										1								
$s = 7$								0										1							
$s = 8$									0										1						
$s = 9$										0										1					

Fig.3

Representation of Information (Exa. 1)

Exa. 1: Standard Binary System

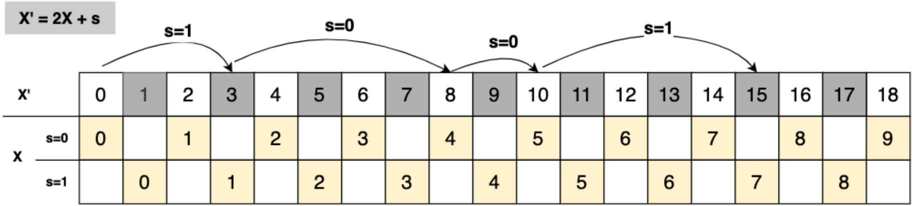

Fig.4

Exa. 1: Standard Binary System

$X' = 2X + s$

s=1 s=0 s=0 s=1

X'		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
X	s=0	0		1		2		3		4		5		6		7		8		9
	s=1		0		1		2		3		4		5		6		7		8	

Exa. 2: Standard Decimal System

$X' = 10X + s$

s=2 s=0 s=2 s=5

N	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		202		2025
s=0	0										1										2				
s=1		0										1													
s=2			0										1												
s=3				0										1											
s=4					0										1							...		...	
s=5						0										1									
s=6							0										1								
s=7								0										1							
s=8									0										1						
s=9										0										1					

Fig.5

Exa. 1: Standard Binary System

X' = 2X + s

X'	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
X s=0	0		1		2		3		4		5		6		7		8		9
X s=1		0		1		2		3		4		5		6		7		8	

Exa. 2: Standard Decimal System

X' = 10X + s

N	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		202		2025
s=0	0										1										2				
s=1		0										1													
s=2			0										1												
s=3				0										1											
s=4					0										1							...		...	
s=5						0										1									
s=6							0										1								
s=7								0										1							
s=8									0										1						
s=9										0										1					

Fig.6

Exa. 3: Asymmetric Binary System

- Let $\mathcal{A} = \{0, 1\}$
- $Pr(0) = 1/4, Pr(1) = 3/4$

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- Let $\mathcal{A} = \{0, 1\}$
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- a cycle of length 4, which contains 1 zero and 3 ones
- $x' = C(s, x) \approx x/p_s$

Exa. 3: Asymmetric Binary System

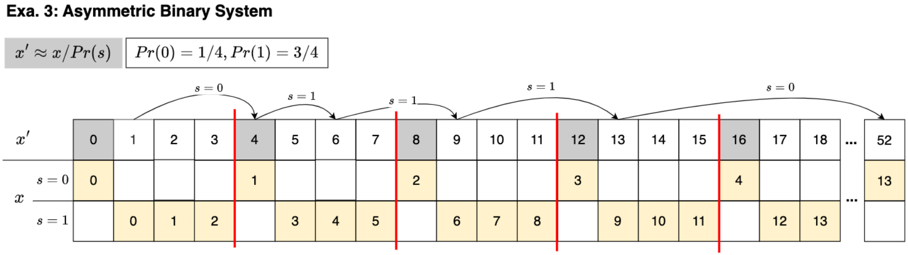

Fig.7

We refine even/odd number by changing their densities

Symmetric or Asymmetric

- Symmetric: uniform distribution
- Asymmetric: non-uniform distribution

tANS

- d

Performance

- f

rANS

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rANS

- ”r” = ”range”: similar to range coding (RC)
- symbol spread function
- representation of information
- asymmetric

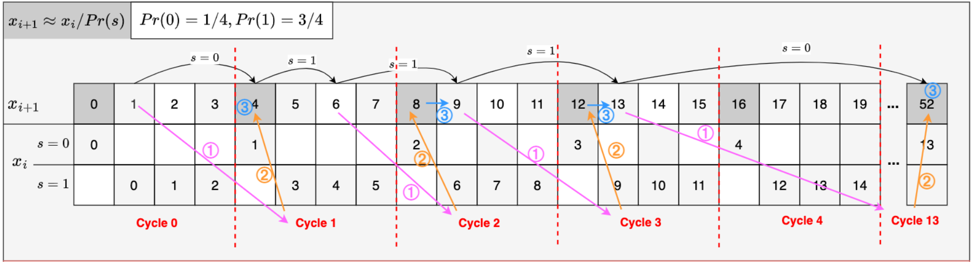
Exa.3

How to encode 101110? What is the encoding function $C(s,x)$?

$x' = C(s,x) \approx x/p_s$

Some notations

- $\mathcal{A} = \{s_1, s_2, \ldots, s_k\}$ is the alphabet set
- s is the symbol, if it is used in subscript: s is the current symbol, $s-1$ is the previous symbol, $s+1$ is the next symbol
- F_s is the absolute frequency of symbol s
- $m = \sum_{s \in \mathcal{A}} F_s$ is the total frequency
- $c_s = \sum_{s \in \mathcal{A}} F_{s-1}$ is the cumulative frequency (c_{s+1} is the cumulative frequency of the next symbol)
- $p_s = F_s/m$ is the probability of symbol s
- x_i is the current state (next state: x_{i+1}, previous state: x_{i-1})



Given current state x_i and next symbol s. How to find next state x_{i+1}, which is the x_i-th appearance of symbol s ?

STEPS / EXAMPLE	$Pr(0) = 1/4, Pr(1) = 3/4$ $s=0$	$Pr(0) = 1/4, Pr(1) = 3/4$ $s=1$	General
Step ①: Find cycle number k	$k = \lfloor x_i/1 \rfloor$	$k = \lfloor x_i/3 \rfloor$	$k = \lfloor x_i/F_s \rfloor$
Step ②: Navigate to the beginning (first number) of that cycle	$4 \cdot k$	$4 \cdot k$	$m \cdot k$
Step ③: Find the position of the symbol s in that cycle - part 1: position of the symbol s in the sequence of the same symbol - part 2: how many other symbols before the symbol s	part 1: first of one $+0$ part 2: no other symbols $+0$	part 1: $+\mathrm{mod}(x_i, 3)$ part 2: only one other symbol (0) and with one of that symbol in one cycle $+1$	part 1: $+\mathrm{mod}(x_i, F_s)$ part 2: only one other symbol (0) and with one of that symbol in one cycle $+c_s$
	$x_{i+1} = m \cdot k$	$x_{i+1} = 4 \cdot \lfloor x_i/3 \rfloor + \mathrm{mod}(x_i, 3) + 1$	$x_{i+1} = m \cdot \lfloor x/F_s \rfloor + \mathrm{mod}(x, F_s) + c_s$

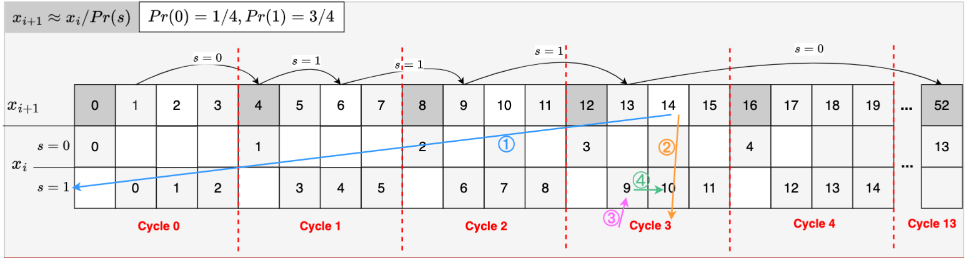
$$x_{i+1} = C(s, x_i) = m \cdot \lfloor x_i/F_s \rfloor + \mathrm{mod}(x_i, F_s) + c_s$$

Fig.8

Encoding Function [2] [5]:

$$x' = m \cdot \lfloor x/F_s \rfloor + \mathrm{mod}(x, F_s) + c_s$$

- **Alphabet set:** $\mathcal{A} = \{a_1, a_2, \ldots, a_k\}$
- **Absolute frequency of symbol** s**:** F_s
- **Total frequency:** $m = \sum_{i=1}^{k} F_i$
- **Probability of symbol** s**:** $p_s = \frac{F_s}{m}$
- **Cumulative frequency before** s**:** $c_s := F_0 + F_1 + \cdots + F_{s-1}$



Given current state x_i. How to find previous state x_{i-1}, where x_i is the x_{i-1}-th appearance of symbol s ?

STEPS / EXAMPLE	$Pr(0) = 1/4, Pr(1) = 3/4$		General
	$s = 0$	$s = 1$	
Step ① : Find the symbol s for the state x_i	if $x_i \bmod 4 = 0$	otherwise or if $x_i \bmod 4 = 1, 2, 3$	s such that $c_s \leq \mathrm{mod}(x_i, m) < c_{s+1}$
Step ②: Find the cycle k of the current state x_i	$k = x_i/4$	$k = \lfloor x_I/4 \rfloor$	$k = \lfloor x_i/m \rfloor$
Step ③: Find the first number of that symbol s in that cycle k	$x_i/4 \cdot 1$	$\lfloor x_i/4 \rfloor \cdot 3$	$\lfloor x_i/m \rfloor \cdot F_s$
Step ④: Allocate to the exact position	$x_{i-1} = x_i/4 \cdot 1 + 0 - 0$	$x_{i-1} = \lfloor x_i/4 \rfloor \cdot 3 + 2 - 1$	$x_{i+1} = \lfloor x_i/m \rfloor \cdot F_s + \mathrm{mod}(x_i, F_s) - c_s$

1. s such that $c_s \leq \mathrm{mod}(x_1, m) < c_{s+1}$
2. $x_{i-1} = D(x_i) = \lfloor x_i/m \rfloor \cdot F_s + \mathrm{mod}(x_i, F_s) - c_s$

Fig.9

Decoding Function (binary):

$$D(x) = (s, F_s \cdot (x >> n) + (x \& mask) - c_s)$$ [2]

```
def rANS_encoder(symbol_counts, s_input):
    # compute cumulative frequencies
    cumul_counts = []
    sum_counts = 0
    for count in symbol_counts:
        cumul_counts.append(sum_counts)
        sum_counts += count
    state = 0
    output = []
    for s in s_input:
        Fs = symbol_counts[s]
        Cs = cumul_counts[s]
        state = (state // Fs) * sum_counts + Cs + (state % Fs)
        output.append((s, state))

    L_avg = math.ceil(math.log(state) / math.log(2.0)) / len(s_input)
    return output, state, L_avg, entropy(symbol_counts)
```

Implemenation in Dash app with interactive visualization

```python
def rANS_decoder(symbol_counts, num_symbols, state):
    cumul_counts = [0]  # Start with 0 as the first cumulative frequency
    total_frequency = sum(symbol_counts)
    for count in symbol_counts:
        cumul_counts.append(cumul_counts[-1] + count)
    def c_inv(y):
        return bisect_right(cumul_counts, y) - 1
    output = []  # List to store decoded symbols
    for _ in range(num_symbols):
        slot = state % total_frequency  # Find the slot in the cumulative frequency range
        s = c_inv(slot)                 # Get the symbol corresponding to the slot
        Fs = symbol_counts[s]           # Frequency of the symbol
        Cs = cumul_counts[s]            # Cumulative count of the symbol
        output.append((s,state))                # Append decoded symbol to the output
        state = (state // total_frequency) * Fs + (slot - Cs)  # Update the state
    return output, state
```

Implementation in Dash app with interactive visualization

Streaming-rANS

Streaming-rANS

To avoid using large number arithmetic, in practice stream variants are used: which enforce by renormalization: sending the least significant bits of x to or from the bitstream (usually L and b are powers of 2). [4]

Example of the problem //TODO: add example

- ANS (Asymmetric Numeral Systems) uses a single integer x, known as the "state," to represent the encoded data.
- Over time, as more symbols are encoded, the state x can grow arbitrarily large. This can lead to:
  - Overflow: If the value exceeds the precision limits of your system (e.g., a 32-bit or 64-bit register).
  - Inefficient Encoding: A larger x means you need more memory or bits to represent it.
- Using the Cfunction from the previous section to encode succeeding symbols, we can encode a sequence into a very large natural number *x*, which can be stored as a length $\lfloor lg(x) \rfloor + 1$ bit sequence. [2]

$[L, H]$

- How to determine L and H?

$[L, H]$

- How to determine L and H?
-

$[L, H]$

- How to determine L and H?
-
-

Method 1: ANS encoding step of symbol sfrom state x [2]

while $x > maxX[s]$ **do** {fff}

  writeDigit(mod(x,b)); $x = \lfloor x/b \rfloor$ {fff}

end while

$x = C(s, x)$ {fff}

Method 2: ANS decoding step from state x [2]

$(s, x) = D(x)$ {fff}

useSymbol(s) {fff}

while $x < L$ **do** {fff}

  $x = b \cdot x + \text{readDigit}()$ {fff}

end while

- Base decoding:
- $x_i = 2 * x_{i-1} + bit$
- Either x_i in $[L, H]$ or x_{i-1} in $[L, H]$. Assume x_{i-1} in $[L, H]$.
  - $x_{i-1} >= L$
  - $x_i >= 2 * L + bit >= 2 * L$
  - x_i can't be in $[L, H]$
  - derive: $H <= 2 * L - 1$
-

Streaming-rANS | choose L and H | Constraint 2: Encoding

```python
for i in range(10):
    print("Hello, World!")
```

Summary

Summary

- To summarize, this may help you to make a Leipzig university beamer presentation.

Reference

Reference

1. C. Shannon, “A mathematical theory of communication,” The Bell Sys- tem Technical Journal, vol. 27, pp. 379–423, 623–656, July, October
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